Shubh Mittal

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RESEARCH INTERESTS

Stellar Evolution, Massive Stars, Transient Astronomy, High-Resolution Spectroscopy, Data Analysis.

EDUCATION

Medium of Instruction: English

Degree	Institute/Board	Grade	Year
Doctoral Student	Astronomical Institute of Czech Academy of Sciences, Charles University	–	2026 - 2030
M.Sc. Physics	Indian Institute of Science Education and Research Thiruvananthapuram	8.77/10	2023 - 2025
B.Sc. Physics	University of Delhi	8.25/10	2020 - 2023

RESEARCH PROJECTS

Search for Luminous Blue Variables in large sky surveys Oct 2026 – 2030
PhD Advisor: Dr. Olga Maryeva *AI CAS Prague, CZ*

- Utilizing astronomical alert brokers and custom machine learning algorithms to discover Luminous Blue Variable (LBV) stars within massive time-domain datasets from ZTF and LSST.
- Member of the OCEANS (Overcoming Challenges in the Evolution and Nature of Massive Stars) project.

Precision Stellar Astrophysics to Probe Cosmic Expansion August 2025 – August 2026
Junior Research Fellow (Project) - Guide: Dr. Anupam Bhardwaj *IUCAA, Pune*

- Performing high-resolution spectroscopy for elemental abundance analysis to study the effect of metallicity on absolute luminosities of Type II Cepheids.
- Using HRS data from SALT telescope, calculated stellar atmosphere parameters, metallicity and elemental abundances using stellar atmosphere models in Local Thermodynamic Equilibrium.
- Developing the Stellar Atmosphere Parameter Estimation module and preparing the entire pipeline for open-source distribution.

Spectral Analysis and Physical Modelling of Gamma Ray Bursts May 2024 – May 2025
Master Thesis - Guide: Dr. Shabnam Iyyani *IISER TVM*

- Analyzed Fermi Gamma-ray Space Telescope data using the threeML and Naima frameworks to model Inverse Compton scattering and investigate the physical origins of prompt-phase emission.
- Investigated relativistic jet structure and observational signatures through spectral analysis to characterize the behavior of high-energy emission mechanisms in Gamma-Ray Bursts.

Extra-Tidal Region of the Galactic Globular Cluster NGC 5466 June 2023 – August 2023
Internship *Miranda House*

- Identified extra-tidal star candidates and their structure in the galactic globular cluster NGC 5466 using GAIA DR3 data.
- Performed data filteration, proper motion analysis, membership probability determination and isochrone fitting.

SKILLS

Programming: Python, C++, MATLAB, ADQL, WebD
Application Packages: IRAF, DAOPHOT, MOOG, DS9, LaTeX, MS Office
Data Analysis: Pymoog, Scipy, Astropy, Sklearn, threeML, Naima
Operating Systems: Windows, Ubuntu

PUBLICATIONS

1. P. Bordoloi, **S. Mittal** and S. Iyyani, "From Empirical to Physical Model: Direct Fits of Optically Thin Inverse Compton Scattering to Prompt GRB Spectra" - link, published in the Astrophysical Journal.

CONFERENCES & WORKSHOPS

44th meeting of The Astronomical Society of India May 2026
IIT Guwahati *In-person*

- Participated in Rubin LSST workshop and presented a poster at this conference.

School on Stellar and Solar Astronomy Jan 2026
Mar Thoma College, Kerala *In-person*

- Gave a talk on Stellar Spectroscopy and guided a project on "Cosmic Distance Scale with Variable Stars in Gaia Space Mission" to undergraduate students.

Stellar Variability: Taking the Pulse of the Universe Nov 2025
IUCAA, Pune *In-person*

- Participation and LoC membar at the international conference on stellar variability and time-domain astrophysics.

ExCoDOT 2025: Exploring the Cosmos with 3.6-m DOT Oct 2025
ARIES, Nainital *In-person*

- Workshop participation, covering proposal writing, observations, photometric and spectroscopic data reduction of data obtained from 3.6 m Devasthal Optical Telescope.

EXTRA-CURRICULARS

Volunteer, National Science Day, IUCAA 28-February 2026

- Took part in poster and video presentation for the National Science Day. Explained various concepts in observational astronomy to general public.

Coordinator of PARSEC - Astronomy Club of IISER TVM August 2024 – April 2025

- As core member of the Club, I planned, coordinated and assisted in organizing educational outreach events.

President of Neutronics Society, Rajdhani College August 2022 – May 2023

- Managed and coordinated a team of 30+ students. Organized various lectures, inter-college competitions and educational visits for students.

REFERENCES

Dr. Olga Maryeva
Researcher, AI CAS
Email: olga.maryeva@asu.cas.in

Dr. Anupam Bhardwaj
Scientist-E (Assistant Professor)
Email: anupam.bhardwaj@iucaa.in

Dr. Shabnam Iyyani
Assistant Professor
Email: shabnam@iisertvm.ac.in